1 A population of bacteria is increasing by 10% each hour.

Find the percentage increase in the population every 3 hours.

Increase by 10% → ×1.1

1.1 × 1.1 × 1.1 = 1.331

(Ans − 1) × 100 = 33.1

33.1%

(Total for question 1 is 2 marks)

×0.9 ×0.8

2 A shop decreases prices by 10% and then by a further 20%.

Rachel says: "Prices have now decreased by 30%".

Is Rachel correct?
You must show your working.

No.

0.9 × 0.8 = 0.72

(1 − Ans) × 100 = 28

Prices have decreased by 28%

(Total for question 2 is 2 marks)

3 Alex invests some money for 3 years in a savings account.
She gets 4% per annum compound interest.

Alex has £5680.56 at the end of 3 years, work how much she invested.

$$x \times 1.04^3 = 5680.56$$

$$x = \frac{5680.56}{1.04^3}$$

$$= £5050$$

£5050

(Total for question 3 is 3 marks)

4 Bailey invests some money for 5 years in a savings account.
She gets 3% per annum compound interest.

Bailey has £3593.75 at the end of 5 years, work how much she invested.

$$x \times 1.03^5 = 3593.75$$

$$x = \frac{3593.75}{1.03^5}$$

$$= £3100$$

£3100

(Total for question 4 is 3 marks)

5 Charlie invests £2500 for 3 years in a savings account.
She gets 3% per annum compound interest in the first year, then x% for 2 years.
$\times 1.03$
Charlie has £2705.36 at the end of 3 years, work out the value of x .

Let multiplier = y

$$2500 \times 1.03 \times y^2 = 2705.36$$

$$y^2 = \frac{2705.36}{2500 \times 1.03}$$

$$y^2 = 1.0506\ldots$$

$$y = \sqrt{\text{Ans}}$$

$$= 1.025$$

$$x = (\text{Ans} - 1) \times 100$$

$$= 2.5\%$$

2.5%

(Total for question 5 is 4 marks)

6 Dana invests £5000 for 4 years in a savings account.
She gets 2% per annum compound interest in the first year, then x% for 3 years.

Dana has £5508.23 at the end of 4 years, work out the value of x .

$$5000 \times 1.02 \times y^3 = 5508.23$$

$$y^3 = \frac{5508.23}{5000 \times 1.02}$$

$$= 1.080045$$

$$y = 1.026$$

$$x = (\text{Ans} - 1) \times 100$$

$$= 2.6\%$$

2.6%

(Total for question 6 is 4 marks)

7 A population model assumes that the number of people living in Stoverton is increasing by x% each year.

The population is expected to increase by 60% in 6 years, work out the value of x.
Give your answer to 1 decimal place.

$$y^6 = 1.6$$

$$y = \sqrt[6]{1.6}$$

$$y = 1.0814$$

$$x = (\text{Ans} - 1) \times 100$$

$$= 8.1\%$$

8.1%

(Total for question 7 is 3 marks)

8 A car's value is decreasing by x% each year.

The car will half in value in 5 years, work out the value of x.
Give your answer to 1 decimal place.

$$y^5 = 0.5$$

$$y = \sqrt[5]{0.5}$$

$$y = 0.87055$$

$$x = (1 - \text{Ans}) \times 100$$

$$= 12.9\%$$

12.9%

(Total for question 8 is 3 marks)

9 The number of rabbits in a field is increasing by x% each year.

The population is expected to double in 5 years, work out the value of x.
Give your answer to 1 decimal place.

$$y^5 = 2$$

$$y = \sqrt[5]{2}$$

$$= 1.148698...$$

$$x = (\text{Ans} - 1) \times 100$$

$$= 14.9\%$$

14.9%

(Total for question 9 is 3 marks)

10 A car's value is decreasing by x% each year.

The car's value will decrease by 60% in 6 years, work out the value of x.
Give your answer to 2 decimal places.

$$y^6 = 0.4$$

$$y = \sqrt[6]{0.4}$$

$$= 0.85837$$

$$x = (1 - \text{Ans}) \times 100$$

$$= 14.16\%$$

14.16%

(Total for question 10 is 3 marks)

11 A circle's radius is increased by 8%.
Find the increase in the circle's area.

$A = \pi r^2$

New circle : $A = \pi (1.08r)^2$

$= 1.1664 \pi r^2$

$(1.1664 - 1) \times 100 = \underline{16.64\%}$

16.64%

(Total for question 11 is 4 marks)

12 A circle's area is increased by 5%.
Find the increase in the circle's radius.

Give your answer to 1 decimal place.

$A = \pi r^2$

$1.05A = \pi (xr)^2$

$1.05A = x^2 \pi r^2$

$1.05 = x^2$

$x = 1.024695\ldots$

$(\text{Ans} - 1) \times 100 = 2.5\%$

2.5%

(Total for question 12 is 4 marks)

13 A cube's length is increased by 12%.
Find the increase in the cube's volume.

Give your answer to 3 significant figures.

$V = l^3$

New cube: $V = (1.12l)^3$

$= 1.404928l^3$

$(1.404928 - 1) \times 100 = 40.5\%$ (3sf)

40.5%

(Total for question 13 is 4 marks)

14 A cube's volume is increased by 25%.
Find the increase in the cube's length.

Give your answer to 3 significant figures.

$1.25V = (xl)^3$

$1.25V = x^3 l^3$

$1.25 = x^3$

$x = 1.077217\ldots$

$(\text{Ans} - 1) \times 100 = 7.72\%$ (3sf)

7.72%

(Total for question 14 is 4 marks)